

Table 1. Full L-CiteEval length-transfer results for Qwen2.5-3B-Instruct. Values are full-sample means $\pm$ ten-group matched-question jackknife standard errors. The 8K, 16K, and 32K conditions use the same 40 question identities. Higher is better for every column.

Context	Method	Answer recall (%) $\uparrow$	Citation F1 (%) $\uparrow$	Concept relevance (0–2) $\uparrow$	Instruction relevance (0–2) $\uparrow$	Fluency (0–2) $\uparrow$
8K	Original	66.2 $\pm$ 8.1	7.8 $\pm$ 1.6	0.00 $\pm$ 0.00	1.35 $\pm$ 0.14	1.50 $\pm$ 0.11
	S-PID	72.1 $\pm$ 9.3	7.2 $\pm$ 2.1	0.00 $\pm$ 0.00	1.60 $\pm$ 0.11	1.50 $\pm$ 0.07
	A-LQR	70.1 $\pm$ 8.9	5.6 $\pm$ 2.0	0.05 $\pm$ 0.05	1.48 $\pm$ 0.11	1.48 $\pm$ 0.09
	H_∞ (ours)	66.4 $\pm$ 7.9	6.4 $\pm$ 2.0	0.00 $\pm$ 0.00	1.55 $\pm$ 0.09	1.55 $\pm$ 0.08
16K	Original	65.2 $\pm$ 7.6	5.8 $\pm$ 2.2	0.20 $\pm$ 0.08	1.40 $\pm$ 0.16	1.52 $\pm$ 0.13
	S-PID	61.9 $\pm$ 10.4	7.2 $\pm$ 3.2	0.00 $\pm$ 0.00	1.38 $\pm$ 0.13	1.48 $\pm$ 0.10
	A-LQR	64.8 $\pm$ 9.8	3.0 $\pm$ 1.5	0.00 $\pm$ 0.00	1.43 $\pm$ 0.15	1.57 $\pm$ 0.11
	H_∞ (ours)	65.2 $\pm$ 8.6	7.8 $\pm$ 3.6	0.15 $\pm$ 0.08	1.40 $\pm$ 0.12	1.43 $\pm$ 0.09
32K	Original	54.0 $\pm$ 8.4	2.4 $\pm$ 1.3	0.00 $\pm$ 0.00	1.18 $\pm$ 0.17	1.48 $\pm$ 0.11
	S-PID	61.0 $\pm$ 9.1	1.6 $\pm$ 1.1	0.00 $\pm$ 0.00	1.20 $\pm$ 0.12	1.35 $\pm$ 0.09
	A-LQR	56.2 $\pm$ 6.7	4.1 $\pm$ 1.9	0.00 $\pm$ 0.00	1.23 $\pm$ 0.13	1.35 $\pm$ 0.08
	H_∞ (ours)	49.2 $\pm$ 6.1	4.3 $\pm$ 2.4	0.00 $\pm$ 0.00	1.18 $\pm$ 0.15	1.20 $\pm$ 0.15